Eric Shan

+64 21 082 13646 | eshan88899@gmail.com | www.ericshan.nz | linkedin.com/in/shan-eric

Education

University of Auckland 2022 – 2026
Bachelor of Engineering (Honours), Electrical and Electronics Engineering *Auckland, New Zealand*

Albany Senior High School 2019 – 2021
NCEA Level 3, Academic Excellence Certificate *Auckland, New Zealand*

Work Experience

Electrical Engineer Intern Oct 2025 – Mar 2026
Progressive Technologies NZ Ltd *Auckland*

- Supported electrical design work, including component selection, schematic development and review of circuit functionality against project requirements.
- Ran circuit simulations to validate designs across operating conditions, reducing risk before hardware implementation.

Electrical Engineer Intern Nov 2024 – Feb 2025
Future Manufacturing Ltd *Auckland*

- Supported mass production of DC-DC converters, including PCB assembly, component placement, soldering and testing.
- Created and maintained schematics, PCB layouts, bills of materials (BOMs) and technical reports.

Projects

Multistatic Radar (Final Year Project) | *ESP32, Wi-Fi CSI, PCB Design* Mar 2026 – Oct 2026

- Designed and built a multistatic radar system using ESP32 microcontrollers and Wi-Fi Channel State Information to detect movement and objects.
- Delivered a working system at a cost of NZ$22.75 per node.
- Rated best project in the Radio Systems category at the University of Auckland Display Day.

Battery Level Monitoring System | *KiCad, PCB Fabrication* Dec 2025 – Jan 2026

- Designed a 12V lithium battery level monitor with LED indicators and an emergency relay cutoff for low-voltage protection.
- Completed the full PCB process: schematic design, circuit simulation, layout, fabrication and assembly.

Energy Monitoring System | *LTspice, Altium Designer, Embedded Systems* Mar 2025 – Jun 2025

- Designed and prototyped a wireless energy monitor using voltage and current sensors for precise power measurement.
- Simulated the circuit in LTspice, developed the PCB in Altium Designer, and assembled it with through-hole and SMD soldering.

DC-DC Flyback Converter | *UC3843, Power Electronics* Jun 2024 – Oct 2024

- Designed and tested a current-mode controlled flyback converter using the UC3843 controller for isolated DC-DC conversion.
- Specified the transformer, selected protection components and verified performance under varying loads.

Water Level Detection System | *FEMM, LTspice, PCB Fabrication* Mar 2024 – May 2024

- Engineered a water level probe using electromagnetic simulation in FEMM, with transmitter and receiver circuits designed in LTspice and fabricated on a through-hole PCB.

Custom FPV Drone | *Soldering, Microcontroller Integration* Dec 2024 – Jan 2025

- Built a quadcopter integrating brushless motors, an ELRS receiver and a flight controller, with safe power management tested under load.

Emergency & Outdoor Lighting Design | *DIALux Evo* May 2024 – Jun 2024

- Created compliant emergency and outdoor lighting layouts for a two-storey building (AS/NZS 2293).

Technical Skills

Engineering Tools	Altium Designer, KiCad, LTspice, PLECS, Autodesk Fusion, DIALux Evo, FEMM
Electronics	Through-hole and surface-mount soldering, Raspberry Pi, ESP32, Arduino
Programming	MATLAB, C/C++, Java, Python
Other Tools	Git, LaTeX, HTML, CSS
Professional	Report writing, technical documentation
Spoken Languages	English (fluent), Mandarin Chinese (fluent)